

Recursive Genomic Q-Former: Parameter-Reduction vs. Performance Trade-off

Task details	
Task	cancer_type — 4-way tumor-of-origin classification from bulk gene-expression profiles (TCGA RNA-seq).
Dataset	Unified TCGA cohort: 2,738 samples $\times$ 20,530 genes; top 4,000 highly-variable genes used as input tokens.
Classes	Breast (BRCA, 867), Head-neck (HNSC, 423), Lung (LUNG, 943), Prostate (PRAD, 505).
Split	70/15/15 train/val/test, 523 test samples , seed 42.
Model	Recursive Genomic Q-Former: $d_{\text{model}} = 128$, 256 marker query slots, recursion depth 4.
Metric	Test accuracy and macro-averaged F1.

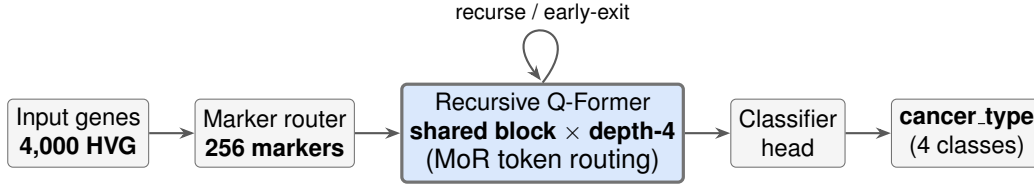


Figure 1: Pipeline. The shared transformer block (blue) is reused across recursion steps; MoR routes tokens to a per-token depth instead of a fixed pass.

All results are on the **cancer_type** classification head (4 classes: BRCA, HNSC, LUNG, PRAD; 523 test samples), with a fixed recursion depth of 4, $d_{\text{model}} = 128$, 256 marker query slots, and seed 42. The “regular” recursive model shares one transformer block across all recursion steps; the comparison models either un-tie those weights (*Independent*) or add token-adaptive routing (*MoR*).

Table 1: Parameters and performance for all three model types. One view of the full trade-off: parameter budget (recurrent-block params, total params, and the independent/shared ratio) alongside all three performance metrics (accuracy, macro-F1, weighted-F1). Sharing the transformer block cuts recurrent params by $\sim 4\times$ for a 0.57-point accuracy cost; Mixture-of-R recursions (MoR) keeps that small budget yet recovers the gap to 0.19 points and is best on macro-F1.

Model	Weights	Routing	Parameters			Performance		
			Recurrent	Total	Ratio	Acc.	Macro-F1	Wtd-F1
Independent stack	untied	none	530,432	1,109,897	$3.99\times$	0.9904	0.9891	0.9904
Recursive (shared)	shared	none	132,992	712,457	$1.00\times$	0.9847	0.9871	0.9847
MoR (token-adaptive)	shared	token	132,992	712,457	$1.00\times$	0.9885	0.9892	0.9885

Read-out. The shared recursive model uses **74.9% fewer** transformer parameters than the independent stack (132,992 vs. 530,432) for a **0.57-point** accuracy cost. Adding MoR token routing

closes that gap to **0.19 points** at the same small parameter budget, and is the only configuration that also reduces FLOPs.

Table 2: Parameter scaling vs. recursion depth. Shared-weight recursion keeps the transformer parameter count essentially flat as depth grows, whereas an independent stack grows linearly. The reduction factor reaches nearly 8 $\times$ at depth 8.

Recursion depth	Shared params	Independent params	Reduction factor
1	132,608	132,608	1.00 $\times$
2	132,736	265,216	2.00 $\times$
4	132,992	530,432	3.99 $\times$
6	133,248	795,648	5.97 $\times$
8	133,504	1,060,864	7.95 $\times$

Table 3: Compute trade-off (depth-4 variants). Early-exit / adaptive recursion lowers the *effective* mean depth and FLOPs per sample relative to a fixed depth-4 pass, at equal parameter count. MoR achieves the lowest compute and the best F1.

Configuration	Transf. params	Mean depth	MFLOPs/sample	Test acc.
Fixed depth-4 (no early exit)	132,480	4.00	268.4	0.9847
Recursive (shared, early-exit)	132,992	2.75	161.5	0.9847
No recursive refine	132,992	2.75	161.5	0.9771
MoR (token-adaptive)	132,992	2.69	159.1	0.9885

Notes. Param ratio in Table 1 is independent/shared transformer parameters. “Transf. params” counts only the recurrent transformer block (the full model incl. embeddings/heads is $\sim 712\text{K}$ for shared variants, $\sim 1.11\text{M}$ for the independent stack). Compute-saving for the early-exit and MoR variants is $\approx 40\%$ of the fixed depth-4 FLOP budget. Token-reduction at the marker stage additionally compresses 4000 input genes to 256 retained markers (93.6% token reduction) at 98.5% accuracy.

Dataset tasks and per-task results

The model is evaluated on the **unified TCGA benchmark**: 2,738 patient samples, 20,530 genes (top 4,000 highly-variable genes used as input tokens). The same recursive architecture (shared block, depth 4, 256 markers) is trained per task. Six prediction targets span tumor identity, clinical staging, and outcome.

Table 4: Benchmark composition. The six TCGA tasks, their type, number of classes, class distribution, and held-out test size.

Task	Target type	Classes	Class distribution	Test n
cancer_type	Tumor of origin	4	867 / 423 / 943 / 505	523
pathologic_stage	Stage I–IV	4	966 / 886 / 528 / 358	548
pathologic_T	Tumor size T1–T4	4	700 / 1323 / 458 / 257	548
pathologic_N	Nodal status N0–N3	4	1474 / 827 / 380 / 57	548
os_binary	Overall survival	2	407 / 2331	548
tumor_status	Tumor-free / w-tumor	2	2139 / 599	548

Table 5: SMART results across all six tasks (shared recursive model, depth 4, seed 42). “Majority” is the always-predict-largest-class accuracy floor. Tumor-of-origin is near-perfect. Clinical-staging and survival targets are intrinsically hard from bulk expression: on the imbalanced binary tasks (*tumor_status*, *os_binary*) the model trades raw accuracy below the majority floor for a much higher *balanced* macro-F1, while on the 4-class staging tasks it clears the floor on *pathologic_stage*.

Task	Classes	Majority acc.	Test acc.	Macro-F1
cancer_type	4	0.344	0.9847	0.9871
tumor_status	2	0.781	0.6825	0.5827
os_binary	2	0.851	0.7372	0.5696
pathologic_stage	4	0.353	0.5000	0.4504
pathologic_T	4	0.483	0.4580	0.4150
pathologic_N	4	0.538	0.4416	0.3120

Notes. All numbers are single-seed (seed 42) test-set results from `results_genonet/` (staging/outcome tasks) and the `cancer_type` head in `results/main.json`. Majority floors are computed from the class distributions in Table 4. The parameter-reduction trade-off on page 1 is reported on `cancer_type`, the highest-signal task.